

Research Article

What Sources Are Natural Resource Managers Using to Make Decisions?

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ABSTRACT Natural resource management is more effective when it includes collaboration among numerous stakeholders who bring multiple perspectives to the decision-making process. In particular, many individuals and organizations (e.g., The Wildlife Society) have advocated for decisions to be science-based as a means for improving management outcomes. But there has been little research evaluating the sources, science-based or otherwise, used by natural resource agencies to support decision-making at the state-level within the United States. We surveyed state natural resource agency personnel in all 50 states to assess the sources of information used to make management decisions and the factors that influenced selection of the sources. Approximately 60% of respondents reported using ≥ 3 different source types when making recommendations. Existing management plans, the peer-reviewed scientific literature, and expert opinion were the most frequently used sources. Perception of source relevance was the most frequent predictor of source use, and the number of management plans a manager had previously written positively predicted the use of management plans, expert opinion, and personal experiences. Ninety percent of respondents reported that increased use of the peer-reviewed literature would or might improve management recommendations. Aspects of academic and agency cultures influenced which articles or journals were used to make recommendations. Our findings indicate managers support the use of peer-reviewed literature as an important component of management decision-making; however, if the goal is to improve the flow of information between scientists and managers, researchers should publish in journals that are readily accessible to managers and present results in a way that clearly articulates relevance to management decisions. © 2021 The Wildlife Society.

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Effective management of natural resources often requires communication and collaboration among stakeholders including scientists, agency personnel, local residents, hunters, and farmers (Reed 2008, Reed et al. 2009). Among such stakeholders, natural resource managers employed by national or state government agencies are best positioned to identify knowledge gaps that may prevent effective management decision-making because they work directly with natural resources. These direct interactions with resources provide natural resource managers with a wealth of personal experiences, which are often drawn upon to make decisions (Pullin et al. 2004). Research scientists working in the areas of applied ecology, wildlife and fisheries biology, forestry, and related disciplines have the training and professional obligation to produce conclusions based on empirical evidence that can help fill the knowledge gaps identified by agency personnel. Opportunities exist to improve the flow of such information between natural resource managers and research scientists to assist with management objectives

and improve agencies' abilities to address them (Pullin et al. 2004, Cvitanovic et al. 2015, Artelle et al. 2018, Fabian et al. 2019).

There have been a growing number of calls in the literature for natural resource decision-making to be made using scientific evidence instead of, or in addition to, experienced-based approaches (Pullin and Knight 2001, Fazey et al. 2012, Cvitanovic et al. 2015, Gillson et al. 2019). Some existing frameworks allow evidence-based sources to explicitly support decision-making alongside expert opinion or other sources (Gregory et al. 2012). For example, structured decision-making has been widely adopted by some federal natural resource agencies (Szymanski et al. 2009, Williams et al. 2009, Marcot et al. 2012, Briske et al. 2017). Although structured decision-making or other frameworks (Williams 2011) may be used to support decision-making within state natural resource agencies, data highlighting the extent to which such frameworks are employed, or the information sources used to support them, remain limited (Fuller et al. 2020). The Wildlife Society, whose mission is "To inspire, empower, and enable wildlife professionals to sustain wildlife populations and habitats through science-based management and conservation" (The Wildlife

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Society 2021), explicitly acknowledges the importance of scientific evidence as it pertains to wildlife management and conservation. But research has failed to demonstrate widespread acceptance of science-based decisions within natural resource management (Keene and Pullin 2011, Fabian et al. 2019).

What are the impediments preventing what Keene and Pullin (2011) refer to as an effectiveness revolution, which would result from a better integration of science into natural resource management and conservation? The problem is not solely a lack of available research or communication efficiency between scientists and practitioners. Decades of research in fields ranging from science communication to organizational learning suggest the problem is with the way we conceptualize a research-implementation gap (Toomey et al. 2017). Managers and scientists may benefit from considering research-implementation spaces, which re-frames the issue to suggest that research and implementation coexist in an arena of shared societal goals (Toomey et al. 2017). A recent review of the literature on science and practice suggests that few individuals hold the perspective of a shared arena but instead perceive a 1-way knowledge flow from science to practice (Bertuol-Garcia et al. 2018). A better understanding of how practitioners engage with scientists and the scientific literature has potential to move the field of conservation and management forward by identifying and removing obstacles to generating these shared spaces.

Previous surveys of practitioners in the environmental sciences indicated that past experiences and interpersonal communication influence professional decision-making and policy more than consultations with the scientific literature. For example, a survey of management plan compilers in the United Kingdom revealed that most individuals frequently (i.e., always used or usually used) relied on existing management plans (60%) or expert opinion (49%) for support and guidance when compiling new management plans (Pullin et al. 2004); in the same study, published scientific articles were always used or usually used by only 23% of respondents. A similar survey of Swiss professionals working in the forest sector also revealed that articles from scientific journals are infrequently consulted as part of the decision-making process, and that personal experience and discussion with colleagues were considered by most to be more important information sources (Fabian et al. 2019). A number of explanations for these patterns have been offered, including difficulty in accessing scientific material, a lack of relevance in the scientific literature, or a general tendency for people to trust personal perceptions more than newly presented data (Pullin et al. 2004, Cvitanovic et al. 2015, Fabian et al. 2019). A better understanding of the use of science in management, and an assessment of barriers restricting information flow between science and management would offer a path to improve evidence-based decision-making.

In the United States, natural resources are managed jointly by federal and state agencies. State agencies generate State Wildlife Action Plans for ≥ 5 years, which serve as

“blueprints for conserving our nation's fish and wildlife” (Association of Fish and Wildlife Agencies 2019) and are specifically used to map out state funding priorities for species and habitats. These plans tend to focus on a broad array of species and habitats and are designed to identify species of greatest conservation need and develop plans to conserve such species (U.S. Fish and Wildlife Service [USFWS], 2017). Given the wide variety of state agency structures and the diversity of natural resources they are charged with managing, state agency managers are a key focal group if the goal is to improve information flow between research scientists and managers. Further, previous assessments of links between science and management in the United States often failed to directly engage agency managers (Mawdsley et al. 2018).

To promote improved understanding of what influences decisions related to natural resources in the United States, we distributed a questionnaire to a subset of state agency natural resource managers in all 50 states. This questionnaire was intended to address 3 objectives: determine sources of information and the extent they are used when managers develop recommendations for management action, identify factors that influence the use of various sources of information for making recommendations for management action, and identify factors that influence whether respondents believe an increased use of peer-reviewed scientific literature would improve recommendations for management action. Personal experience was demonstrated to be a foundation for decision-making in previously surveyed natural resource focal groups (Pullin et al. 2004, Fabian et al. 2019), so we hypothesized state agencies in the United States would also rely on this source. Objectives 2 and 3 were observational; we pursued these objectives to collect baseline data that might improve collaborations between research scientists and managers.

METHODS

Study Population

We used the State Wildlife Action Plan (SWAP) from each state as a starting point in developing our sample frame. We selected 10 individuals from each of the 50 state wildlife agencies who served in a biologist or manager position within the agency. To determine the 10 people selected from each state, we identified all individuals listed as SWAP authors or contributors (i.e., contributors). We excluded any contributors that were not employees of the state agency from our sample frame (e.g., USFWS, non-governmental organizations). If team leaders were identified by name as contributors within the SWAP, they were given priority in the selection process; however, if >10 team leaders were listed, we randomly selected 10 of them for inclusion in the sample frame. We randomly selected regular team member contributors after we chose all team leaders until we had reached our goal of 10 contributors from each state. In some instances, not enough state employee contributors were listed in the SWAP to meet our goal of 10 for a given state. In such cases, we contacted known SWAP contributors and

asked them to supply a list of other contributors still employed with the state agency, along with their current contact info, and made random selections for their inclusion. If we had no information about known contributors, we contacted the state agency main offices for assistance with that information.

We collected emails for all selected agency biologist or managers and included them in a list as our study sample frame. Although we drew our sample frame primarily from the pool of SWAP contributors, these individuals make many other management decisions outside of the SWAP framework; thus, they offer insight from a broad range of natural resource perspectives. Five of our email messages were never delivered to the intended target; thus, our final sample frame included 495 agency employees in biologist or manager roles from 50 states (i.e., managers).

Survey Instrument

We used the Qualtrics online survey platform (Qualtrics, Provo, UT, USA) to implement our questionnaire (available online in Supporting Information) through email distribution. We sent our questionnaire to 29 state agency managers within the authors' professional network for pre-testing, taking care to exclude anyone who was selected in the study sample frame. We used pretest responses ($n = 4$) to modify and hone the survey prior to study implementation, but we did not use these responses in the analysis. We received approval from the Clemson University Institutional Review Board (IRB 2018-006) to collect data for the study.

We sent out the first communication to our final sample frame of 495 managers on 10 July 2018. Qualtrics' distribution records immediately indicated that 84 of the 495 email addresses were not received and 26 other addresses were undeliverable. We called the agency offices to secure corrections for the 110 email addresses, and subsequently sent out a new email to corrected addresses on 13 July 2018. We sent follow up emails to non-respondents on 24 July and 27 August 2018.

Our questionnaire included sociodemographic questions (i.e., gender, age, race). We also asked respondents to provide their highest level of education completed, the year of completion, and field of study associated with that degree. To better understand the professional responsibilities of respondents, we asked them to indicate how often their current position includes duties related to administration, game management, habitat management, non-game management, policy, public outreach or education, research, species eradication, and species recovery (7-point Likert scale, where 1 = not at all, 4 = once/month, and 7 = daily).

Our questionnaire also included questions about the nature of recommendations for management action made by respondents. We defined management actions in our questionnaire as any attempt to control or manipulate factors related to natural resources. We simultaneously defined and inquired about the format of recommendations for management actions (i.e., recommendations) made within the respondent's agency in a question with the following

answer options: management plans, informal recommendations for management action (e.g., verbal agreement, directive), and written recommendations for management action (e.g., memo, short document, email). Henceforth, we refer to the term recommendations for management action as recommendations.

We asked respondents several questions designed to help us understand their experiences in their respective agency and the workflow within the agency when developing recommendations. We asked respondents if the answers provided represented their personal perspective, their agency's perspective, or both. We then asked respondents to describe the process of how recommendations were developed and approved in their agency (open-ended question), which we qualitatively analyzed for themes. We also solicited information on how frequently internal and external collaborations took place for recommendations, and how many collaborators, including themselves, were typically involved in the recommendation development process at their current agency. We asked respondents to indicate an approximate number of recommendations they had contributed to in their career (response category options were 0, 1–5, 6–10, with >41 as the maximum response option). We also asked what types of recommendations they had developed in the past (e.g., SWAP, species management). We presented respondents with a list of potential information sources (e.g., government publications, management plans, peer-reviewed scientific literature), and requested respondents indicate which sources they use to assist with recommendations and how frequently they use each specific source in the development of recommendation (7-point Likert scale, where 1 = never, 4 = sometimes, and 7 = always). The source options were based on previous research (Pullin et al. 2004) and our own experience in the field; however, we presented respondents with another category (i.e., other) in the event an important option was missing from our list. Similarly, we asked what factors influence their choice of sources used when developing recommendations (e.g., agency policy, perception of source credibility, the relevance the source has to recommendations; 7-point Likert scale, where 1 = strongly agree, 4 = neutral, and 7 = strongly agree).

We posed several questions specifically focused on respondents' use of peer-reviewed scientific literature in recommendations. We asked how many peer-reviewed scientific articles they typically reference in recommendations, and how many peer-reviewed articles they had authored or co-authored. We also solicited information on the degree to which certain factors were relevant when selecting journals or specific articles in the peer-reviewed literature for use in recommendations. We asked respondents to consider factors such as their agency's policies, the political climate, impact factor, the journal's relevance to their recommendation topic, and access to the journal (i.e., cost of the journal, ease of accessing the journal, successful past access to the journal, time required to access the journal). We asked respondents whether they thought an increase in the use of peer-reviewed scientific literature would improve their recommendations. Finally, we asked respondents to

list up to 3 peer-reviewed scientific journals they find useful when developing recommendations (presented as an open-ended question).

Data Analysis

We used descriptive statistics to determine the sources of information and the extent those sources are used when managers develop recommendations. To determine factors that influence the type and frequency of information used in recommendations, each of our sources (i.e., government publications, management plans, opinion [shared expertise], opinion [complementary expertise], peer-reviewed scientific literature, personal experience, public opinion) served as dependent variables in 7 separate linear regression models (1 for each source) with 7 additional variables serving as our predictors, with 1 exception described below (Table 1). We modified 2 of the predictor variables; first, we collapsed the answer options related to recommendation format to create a binary variable (those managers that used only formal management plans vs. all other possible responses; Table 1). Second, we combined 4 variables related to access (i.e., ease of access, cost of access, time required to access, successful past access to the source) into a single access variable by taking the mean response across the 4 access factors. We made the decision to combine the variables after confirmatory factor analysis of the 4 variables revealed a single latent variable model had more support ($P=0.48$, $df=2$) than a null model ($P<0.05$, $df=6$) with no covariance structure (P -values are from χ^2 test of null hypothesis that a given model fits the data). We further assessed the internal consistency of this combined variable using Cronbach's alpha ($\alpha=0.79$). We did not include the access variable in the model with personal experience as a response variable because access to one's own personal experience is a given. We standardized continuous predictors to z-scores prior to analysis.

We used multiple logistic regression to determine which factors influenced whether respondents thought an increased use of peer-reviewed scientific literature could

improve recommendations. For our dependent variable, we collapsed the maybe and no responses into a single response (i.e., skeptics) to compare against the yes responses (i.e., advocates). We used 5 independent variables in our model (Table 2) and standardized continuous predictors to z-scores prior to analysis.

We estimated non-response bias using the wave analysis methodology (Armstrong and Overton 1977). We compared responses from 12 questions for the entirety of the final (i.e., third) wave ($n=22$) against the first 22 responses received (first wave) for 9 variables (Tables 1 and 2). We treated recommendation of the format, number of career recommendations, whether respondents believed the use of peer-reviewed scientific literature would improve recommendations (skeptics), number of peer-reviewed scientific articles authored, education, and gender as categorical and compared responses using a chi-square test of independence. Age of respondent and years since graduation followed a normal distribution and exhibited equal variance around the mean, so we used a Student's t -test to compare these variables. The use of peer-reviewed literature, influence of agency policy, source relevance, and access to the source did not follow a normal distribution (Shapiro-Wilk; all $P<0.05$), so we compared first and third wave responses using the non-parametric Mann-Whitney U test. We used SPSS (IBM, Armonk, NY, USA) or R (version 4.0; R Core Team 2020) to calculate all descriptive and inferential statistics.

RESULTS

Sample Description

We received 166 responses to our questionnaire, and 137 were usable (27.6% response rate). We received ≥ 1 response from all but 5 states; we did not receive any responses from California, Hawaii, Nevada, New Hampshire, or Wisconsin employees. Our test for non-response bias revealed no significant difference between first and third wave responses (all P -values < 0.05).

Table 1. Summary of independent variables used in a linear regression model to predict frequency of use for various sources of information in the development of recommendations from a survey of state wildlife agency biologists and managers from 45 states in the United States, July–August 2018.

Variable	Description	$\bar{x}$ or %	SD
Recommendation format	0 = informal and written recommendations for management action; 1 = management plans	0 = 88.1% 1 = 11.9%	
Career recommendations	Number of recommendations made throughout career (1 = 0; 2 = 1–5; 3 = 6–10; 4 = 11–15; ...; 7 = 26–30; 8 = 31–35; 9 = 36–40; 10 = ≥ 41)	6.9	2.9
Agency policy	Extent of influence of agency policy in the selection of data and information sources (7-point Likert scale where 1 = not influential at all, 4 = somewhat influential, and 7 = extremely influential)	5.4	1.4
Source relevance	Extent of influence of relevance of source in the selection of data and information sources (7-point Likert scale where 1 = not influential at all, 4 = somewhat influential, and 7 = extremely influential)	6.2	0.9
Access	Mean of responses to extent of influence of 4 access-related variables (cost, ease, past, and time required to access) in the selection of data and information sources (7-point Likert scale where 1 = not influential at all, 4 = somewhat influential, and 7 = extremely influential); combined variable	5.0	1.0
PRSL ^a use	Number of peer-reviewed scientific articles referenced in recommendations (1 = 0; 2 = 1–5; 3 = 6–10; 4 = 11–15; 5 = ≥ 16)	2.5	1.2
Education	Highest level of completed education; 1 = high school; 2 = vocational or technical; 3 = some college (no degree); 4 = associate's degree; 5 = bachelor's degree; 6 = some graduate study (no degree); 7 = master's degree; 8 = doctoral (Ph.D., MD, DVM, JD)	6.9	0.9

^a PRSL = Peer-reviewed scientific literature.

Table 2. Predictor variables used in a logistic regression model to evaluate the factors that influenced respondent's opinions as to whether increased use of peer-reviewed scientific literature would improve recommendations for management action from a survey of state wildlife agency biologists and managers from 45 states in the United States, July–August 2018.

Variable	Description	$\bar{x}$	SD
Career recommendations	Number of recommendations made throughout career (1 = 0; 2 = 1–5; 3 = 6–10; 4 = 11–15; ...; 7 = 26–30; 8 = 31–35; 9 = 36–40; 10 = ≥ 41)	6.9	2.9
PRSL ^a use	Frequency of use of peer-reviewed scientific literature to support recommendations (7-point Likert scale where 1 = never, 4 = sometimes, and 7 = always)	5.5	1.2
PRSL ^a authored	Number of peer-reviewed scientific publications authored or co-authored (1 = 0; 2 = 1–5; 3 = 6–10; 4 = 11–15; 5 = ≥ 16)	2.5	1.2
Education	Highest level of completed education; 1 = high school; 2 = vocational or technical; 3 = some college (no degree); 4 = associates degree; 5 = bachelor's degree; 6 = some graduate study (no degree); 7 = master's degree; 8 = doctoral (Ph.D., MD, DVM, JD)	6.9	0.9
Years since graduation	Number of years since graduated with highest level of education (fill in the blank)	19.9	9.1

^a PRSL = Peer-reviewed scientific literature.

The mean age of respondents was 49 ± 9.28 (SD) years; 66% were female, and 97.5% self-identified their race as white. The majority in our sample (78.7%) held a master's degree or higher in a topic related to wildlife, and the median year of completion of the highest degree was 1999 (19 years prior to completing the survey). Most (97.6%) indicated that their duties predominately involved management or conservation of natural resources (e.g., game, non-game, or habitat management, species recovery, invasive species eradication). Ninety-eight percent of respondents indicated they were answering the questions based on both their personal and professional perspective, or only their professional perspective, which indicates these answers largely reflect agency views and not idiosyncratic feedback.

Management Recommendation Overview

The majority of respondents (62.2%) indicated their agencies typically use all 3 formats of recommendations for management action; others (14.4%) indicated use of written format types (i.e., management plans and written recommendations) without the use of informal verbal directives. Our results suggest that states have a variety of approaches for approving management actions depending on the scale of the project; per qualitative coding of narrative responses, we noted a general trend where smaller management scale or scope activities (e.g., local-scale manipulations) could be made by individual managers, but larger scale activities (e.g., surveys, bag limits, species listing, general guidelines for species management) were typically approved by a mid-level administrator (e.g., division chief) or higher. Most respondents indicated they had made recommendations for habitat management (67%) and SWAP development (55%). Forty-three percent had been involved in non-game management recommendations, and another 34% had made recommendations related to harvest (e.g., game, timber, fisheries). Respondents tended to be experienced with recommendations; 54.3% had contributed to ≥ 41 recommendations (the highest response category) for management actions for their current employer, and 61% had contributed to ≥ 41 during their career. Greater than 80% of respondents indicated ≥ 4 people (3 others and themselves) collaboratively contributed to recommendations; only 2 respondents indicated recommendations were not

developed in collaboration with others. Respondents indicated internal collaborations ($\bar{x} = 6.1 \pm 1.2$) were more common than collaborations external to their agency ($\bar{x} = 4.8 \pm 1.5$).

Sources of Information and Reasons for Selection

Respondents used of a variety of sources of data and information to aid the development of their recommendations (Fig. 1). Our respondents reported usually or always using a mean of 3 different sources for the development of their recommendation. Personal experience (59%), the peer-reviewed scientific literature (58%), and expert opinion from others with similar background (51%) were the most frequently used source types. Selection of a given source was most influenced by state and federal laws and regulations ($\bar{x} = 6.3 \pm 1.1$) and perception of source relevance to the topic ($\bar{x} = 6.2 \pm 0.9$), whereas selection of sources was least influenced by political climate ($\bar{x} = 3.8 \pm 1.5$; Fig. 2).

With respect to peer-reviewed scientific literature specifically, 82% of respondents had published ≥ 1 peer-reviewed scientific article, and 63% of respondents used 1–5 peer-reviewed articles when developing a recommendation. Multiple sections of a scientific paper were considered most useful when developing recommendations, but the following sections appeared to be particularly useful: management implications (82%), conclusions and discussion (74%), abstract (65%), and results (57%). Source relevance to topic was the most influential determinant for their selection of peer-reviewed literature for inclusion in recommendations ($\bar{x} = 6.4 \pm 0.8$; Fig. 2). Similar to the result obtained for all sources, political climate was the least influential consideration for selection of peer-reviewed literature when developing recommendations. The specific journals cited most often within our sample as useful for developing recommendations were *Journal of Wildlife Management* (19% of responses provided) and *Conservation Biology* (8%). Taxa-specific journals also were reported as useful for recommendation development. When pooled within taxonomic groupings, journals focused on fisheries (11%), herpetology (8%), birds (5%), and mammals (4%), collectively represented 28% of responses provided. Most of the remaining responses were journals focused on natural history, general ecology, and habitat management (aquatic

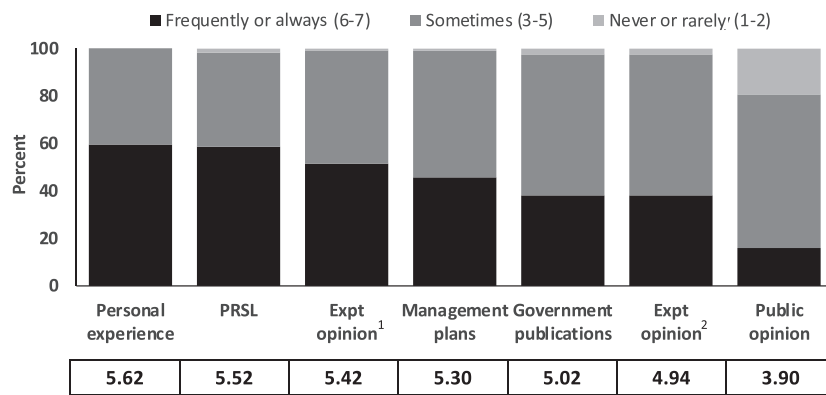


Figure 1. Percent of respondents that never or rarely (1–2), sometimes (3–5), or frequently or always (6–7) use each of 7 possible sources to support recommendations for management action. Results are from a survey of state wildlife agency biologists and managers from 45 states in the United States, July–August 2018. Values in the table below the graph are the overall mean Likert score for each source. PRSL = peer-reviewed scientific literature; Expt opinion¹ = expert opinions from individuals that share the respondent's expertise; Expt opinion² = expert opinions from individuals that have complementary expertise relative to the respondent.

and terrestrial). Respondents provided 68 unique journal titles.

The majority of respondents indicated that the increased use of peer-reviewed scientific literature would (44%) or might (46%) improve their recommendations. Respondents indicated peer-reviewed literature could be more readily integrated into recommendations if changes were to occur within agencies and among those conducting and publishing research. The agency-related changes respondents viewed as the most influential in facilitating increased use of peer-reviewed literature were better access to journals ($\bar{x} = 5.8 \pm 1.4$) and more time allotted to developing the recommendation ($\bar{x} = 6.0 \pm 1.2$); additional training in complementary disciplines was seen as the least influential agency change ($\bar{x} = 4.6 \pm 1.4$). The research-focused change

viewed as most likely to promote the use of peer-reviewed literature in agency recommendations was production of more-relevant scientific literature ($\bar{x} = 5.5 \pm 1.4$). Minimizing unnecessary jargon ($\bar{x} = 4.7 \pm 1.3$) and article length ($\bar{x} = 4.2 \pm 1.3$) were scored as research changes least likely to influence use of the peer-reviewed literature in agency recommendations.

Models constructed to predict the extent to which sources were used in recommendations all had weak predictive power (Table 3). Across models, source relevance was the most consistently significant predictor. It was positively associated with the use of management plans, peer-reviewed scientific literature, personal experience, and public opinion. The number of management recommendations a respondent had made throughout their career was a significant

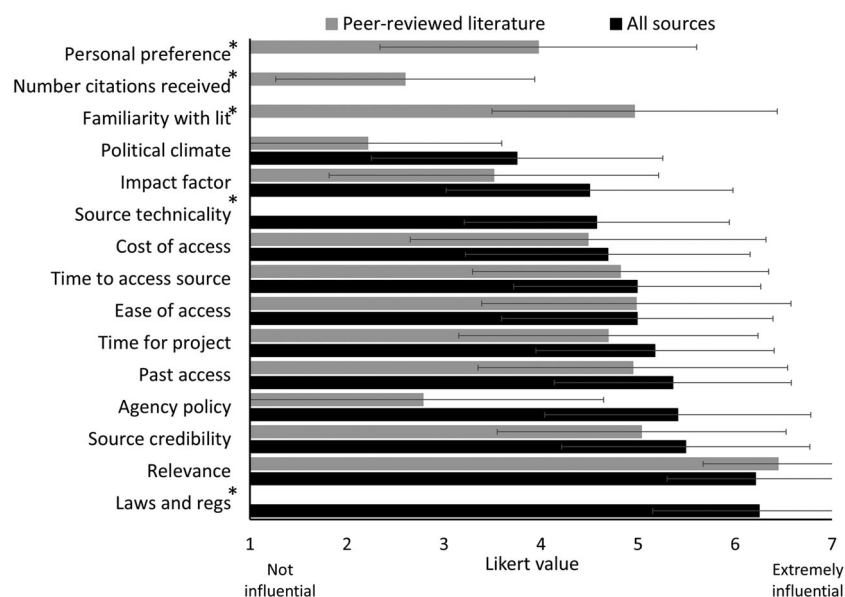


Figure 2. Mean influence of several factors (y-axis) on the selection of all sources (black bars) and peer-reviewed scientific literature (gray bars) when state-agency biologists made recommendations for management action. Survey results are from 45 states in the United States, July–August 2018. Response options differed between the question for all sources and peer-reviewed scientific literature; the absence of a bar for responses marked with an asterisk indicates the category was not presented as an option for that source. Error bars represent 1 standard deviation above and below the mean. Lit = literature.

Table 3. Linear regression model results for predictors of information source used when developing recommendations for management action from a survey of state wildlife agency biologists and managers from 45 states in the United States, July–August 2018. Columns represent the response variable (i.e., frequency of use of information source for recommendation) for each of the 7 models and the associated standard error (SE). Rows represent hypothesized predictors of the extent of use for a source as measured by a 7-point Likert scale.

Potential predictors	Sources of information													
	Government publications		Management plans		Opinion (shared expertise) ^a		Opinion (comp expertise) ^a		PRSL ^b		Personal experience		Public opinion	
	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE
Recommendation format	0.83*	0.38	0.69*	0.32	0.00	0.32	0.07	0.37	0.45	0.35	0.17	0.35	0.38	0.42
Career recommendations	0.17	0.12	0.20*	0.10	0.03	0.10	0.25*	0.12	0.0	0.12	0.26*	0.12	−0.05	0.14
Agency policy	−0.03	0.12	−0.08	0.10	0.12	0.10	−0.09	0.12	−0.07	0.11	0.00	0.11	0.13	0.13
Source relevance	0.14	0.12	0.47***	0.10	0.17	0.10	0.06	0.12	0.26*	0.11	0.34**	0.11	0.35*	0.14
Access	0.21	0.12	0.03	0.11	−0.04	0.11	0.16	0.13	0.11	0.12			−0.13	0.14
PRSL ^a authored	0.19	0.13	−0.16	0.11	−0.19	0.11	−0.19	0.13	0.21	0.12	0.00	0.12	−0.25	0.14
Education	−0.06	0.13	−0.28*	0.11	−0.01	0.11	−0.02	0.13	0.11	0.12	−0.02	0.12	−0.06	0.15
Model fit (r^2)	0.11		0.28		0.07		0.08		0.13		0.16		0.13	

* $P \leq 0.05$, ** $P \leq 0.005$, *** $P \leq 0.0005$.

^a Opinion was divided into the opinion of someone who shares the respondent's expertise, and the opinion of someone who has complementary (comp) expertise.

^b Peer-reviewed scientific literature.

positive predictor for 3 sources: management plans, the opinion of colleagues with complementary (non-overlapping) expertise, and personal experience. Level of education was negatively correlated with the use of management plans as a source but unrelated to use of the peer-reviewed literature or any other source.

Respondents who were already more likely to use peer-reviewed literature as a source and who graduated more recently were significantly more likely to express confidence that an increased use of the peer-reviewed literature would improve recommendations (Table 4). For every unit increase on the Likert scale response to our question about the frequency of peer-reviewed literature use in management recommendations, respondents were 2.5 times more likely to feel the use of scientific literature would improve recommendations. For every increase in the year since graduation, respondents were less likely to be an advocate for use of peer-reviewed literature by a factor of 0.49.

DISCUSSION

To increase rigor in wildlife decision-making, it is important to understand what sources of information are used in natural resource recommendations (Pullin et al. 2004, Fuller et al. 2020). Nearly all respondents in our sample of state agencies indicated a broad use of source types when making recommendations for management action related to natural resources. We assume the breadth we identified exists because the need to find relevant material to guide decision-making is perceived to be more important than the source itself. Most individuals were open to the idea that increased use of the peer-reviewed literature would improve management recommendations and indicated that barriers to increased use were present within research and agency cultures.

The peer-reviewed scientific literature provides a source of evidence that is viewed as credible, yet respondents in our survey were no more likely to use this source than several others we tested. Our findings suggest the peer-reviewed literature is used by state agency natural resource managers in the United States slightly more than managers in other countries (Pullin et al. 2004, Fabian et al. 2019). Nevertheless, our results suggest that respondents viewed the peer-reviewed literature as a source that may not be a necessary component to decision-making but, when used, is often paired with a second source.

Managers who had been out of school for longer periods of time appeared to be more skeptical about the value of the peer-reviewed literature as it pertained to management recommendations, which is counter to results from the medical field (Schulz et al. 2013). Skeptical respondents to our survey may believe that years of experience in the field offer better insights into the management options than more recently published research on the topic. This possible explanation is bolstered by our finding that individuals who had generated more recommendations for management

Table 4. Results from a binary logistic regression that evaluated predictors of whether respondents felt an increased use of peer-reviewed literature would improve recommendations for management action from a survey of state wildlife agency biologists and managers from 45 states in the United States, July–August 2018. Advocates (yes responses) were coded as 1 and skeptics (i.e., responded maybe or no) as 0 for the response variable.

Predictor	β	SE	Exp(β)
Career recommendations	0.19	0.24	1.21
PRSL ^a use	0.90**	0.27	2.5
PRSL ^a authored	0.51	0.27	1.67
Education	−0.06	0.26	0.94
Years since graduation	−0.72*	0.26	0.49

* $P < 0.01$, ** $P < 0.001$.

^a Peer-reviewed scientific literature.

action were more likely to rely on personal experience. Rupp (2012) suggested that both agency management and academic research are shifting from a species-focus agenda to more ecosystem-level approaches; however, the rate of shift is occurring more rapidly within academic research. If true, the differences in rate of change between these groups could explain why individuals who have been with agencies for an extended time are more skeptical of recent research developments. We broadly defined recommendations for management action to include local and regional actions, so it may be the case that informal sources (e.g., experience, the opinions of others) are entirely sufficient to make minor decisions, whereas formal sources, such as the peer-reviewed literature, are accessed for more complex management problems.

Natural resource managers in our sample selected sources because they were relevant to the decision at hand. It is an open question as to how narrowly focused a scientific journal article must be for natural resource managers to view it as relevant for a particular management decision. Ecological systems and conservation issues often are so complex that managers may not believe they can transfer results across study contexts or scales (e.g., geographic region, sample, timeframe; Roach et al. 2017) or species (Lindenmayer et al. 2015). Nevertheless, researchers are under pressure to generalize their research to such an extent as to increase the likelihood that it is published in a prominent journal and cited frequently (Farwig et al. 2017), 2 factors that can enhance the probability of securing a job and, for current university faculty, tenure and promotion (DeRond and Miller 2005). Therefore, the problem may not be the supply of scientific literature but rather the challenge of developing and using processes, such as adaptive management or collaboration, that more explicitly link the scientific supply of research with management's demands (McNie 2007).

Management plans were among the most frequently used sources by our respondents. Management experience (assessed by the number of career recommendations made) and the respondents' level of education appeared to influence the use of management plans, yet the influence of these variables was opposite in their direction of effect. Managers who have more experience were more likely to use management plans, perhaps because they have more belief in the value of such plans than less-experienced managers. Managers with higher levels of education were less likely to use management plans when making management recommendations, which may be because of their increased exposure to scientific literature during post-graduate research (Hubbard and Dunbar 2017).

The use of relevant management plans and government publications in management decisions would seem to be a logical and natural part of the decision-making process in state agencies. The accessibility of this body of gray literature, however, is likely inconsistent among agencies. Unlike scientific literature that is formally published, gray literature may not be available outside of the originating agency or state; if it is, it may be difficult to find.

Conversely, in-house management plans may be so relevant and trusted that they preclude the incorporation of new information into the management decision-making process.

Aligning with research in the United Kingdom (Pullin et al. 2004) and Switzerland (Fabian et al. 2019), our results suggest that personal experience and expert opinion are consistently used to support natural resource decision-making in the United States. Personal experience and expert opinion offer inexpensive and quick solutions for most problems and are, presumably, the starting point for understanding and determining available options for addressing specific natural resource management issues. Nevertheless, managers and policy makers should be aware of the shortcomings associated with both sources of information. Psychological researchers have reported that decisions from personal experience tend to underestimate the likelihood of rare events, simply because the decision-maker has probably never experienced that event (Hertwig et al. 2004). It is also possible that more experienced managers have a greater breadth of experience to draw upon, and therefore may rely on heuristics, which can sometimes lead to bias (Iftekhhar and Pannell 2015). Whereas experiential knowledge can be a part of conservation decision-making, such decisions may be improved when made in conjunction with other sources of information (Fazey et al. 2006). Expert opinion expands the scope to collective personal experiences across trained individuals. Expert opinions may range from co-workers to supervisors, each with little to decades of experience, and individual decision-makers likely weigh such opinions differently. Nevertheless, even experts in a formal solicitation exercise can introduce biases unless rigorous precautions are put into place, such as iterative testing of elicitation protocols and robust methods for addressing uncertainty (Morgan 2014).

Although public engagement is desirable, and perhaps mandated, within an agency, the difficulties associated with executing the engagement may explain the low rate of use of public opinion we documented. Whether and how to include public opinion in natural resource decision-making is complicated and often debated among researchers and agency personnel (Achillas et al. 2011, Booth and Halseth 2011, Simpfendorfer et al. 2011, Reed et al. 2018). There are resources and methods available for deciding when public input is appropriate and how to access it (Mortenson and Krannich 2001, Lauber et al. 2012, Rodriguez and Peterson 2020), but the decision to include public opinion in natural resource decision-making ultimately varies by agency and topic (Bisbal 2001, and Alberts 2007). In some cases, federal reviews require public input (e.g., National Environmental Policy Act of 1970, Endangered Species Act of 1973); in others, it is up to the discretion of the agency. Public engagement may be particularly important in areas where indigenous knowledge exists that can be integrated with scientific evidence to provide more insight and perspectives on a given topic (Ban et al. 2018). Partnerships between social science researchers and wildlife agencies can make the process of

stakeholder engagement and the translation of results into management recommendations more efficient (Fuller et al. 2020). Our results make clear that public opinion was less frequently referenced than other sources of information when our sample population made management decisions. Future research may help determine the circumstances under which increased reliance on public opinion would bolster or impair natural resource management.

Although we accomplished our objectives, there were limitations of the research that could be addressed in future work. Our survey response rate was low, so the results may include non-response bias. Response rates appear to be decreasing for surveys in recent years (Stedman et al. 2019), but if non-respondents do not differ significantly from respondents, then survey results are valid. Results from the wave analysis did not provide evidence of a difference between early and late respondents, and late respondents can act as a surrogate for those not responding at all (Armstrong and Overton 1977). Our models of factors predicting which information sources were used had low predictive power, so some factors that influence source use may still need to be identified. Our analysis of the variation in management decisions was qualitative, and thus establishes a baseline for future research that may employ quantitative methods. Future research may consider scrutinizing variation among management recommendations; for example, do the information sources shaping decisions differ as spatial scale of the decision or the number of stakeholders increases? Future studies in this area could also survey populations of natural resource managers at the federal level or those employed by non-governmental organizations. Finally, our research focused on the sources used to make management recommendations, but we did not attempt to link sources of information to decision quality or overall satisfaction with final decisions. Identifying information that promotes quality natural resource management decisions would provide important knowledge for all stakeholders concerned with natural resource management.

The relevance of scientific research to management, and its use in decision-making, will almost certainly be increased when managers or agency natural resource managers collaborate directly with researchers. Such collaborations increase the understanding of constraints faced by both groups (e.g., money, time, and available data), and yield a bi-directional flow of information that may bolster decision quality (McNie 2007). Although these collaborations are more likely to influence evidence-based management and conservation, the evidence is only transparent and potentially transferrable to other systems if it is published in accessible literature (Fuller et al. 2020). The publication of management-relevant findings in the gray literature may offer a fast-track alternative that could be viable if accessibility is increased within and outside of agencies (MacDonald et al. 2020). Publishing results in the peer-reviewed literature remains the primary means for maintaining quality and trust among users of information (Haider and Åström 2017).

In summary, appeals for more evidence-based decision-making have been made within the fields of healthcare (Rosswurm and Larrabee 1999), policy (Slavin 2002), and natural resource management (Pullin and Knight 2001). Natural resource managers at state agencies throughout the United States use scientific evidence when making decisions, but they use many other sources as well. Expert opinion is commonly employed, and managers that have been with agencies for longer amounts of time are more skeptical regarding the value of peer-reviewed research; these individuals likely have a wealth of personal experience to draw from and an extensive network of professionals with whom they can consult to forgo reliance on peer-reviewed publications. Both personal experience and the experiences of colleagues are indisputably valuable; however, such sourcing may lead to management strategies that miss emerging threats such as climate change or wildlife disease. Even if such threats are not missed, management that is based on science will aid transparency in decision-making (Fuller et al. 2020), create opportunities for adaptive management, and offer a defensible basis from which controversial management decisions can be conveyed to the public.

MANAGEMENT IMPLICATIONS

Our study offers insights for researchers and state agency personnel involved in natural resource management. Researchers who hope to have their results considered by natural resource managers should develop strategies to promote the relevance of the research to present and future management questions. Such strategies likely include direct collaboration with agency managers and using SWAPs as a guide for developing management-relevant research agendas. The frame developed around a research project and the journal targeted for publication are important considerations for researchers who desire to influence management. When research is framed to appeal to a wide range of readers, it may not be viewed as relevant for more localized decisions. Agencies that want to foster a culture of science-based decisions may need to develop formal mechanisms to encourage the use of the peer-reviewed literature and other sources of evidence among those individuals that have a wealth of experience. There is value in leveraging past experience and expert opinion for decision-making, but these sources of information may be more effective when coupled with scientific assessments that provide information on emerging threats or system-wide dynamics. Promoting The Wildlife Society's (2021) mission of "science-based management and conservation" will require an evaluation of and adjustments to current practices by research scientists and natural resource managers.

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